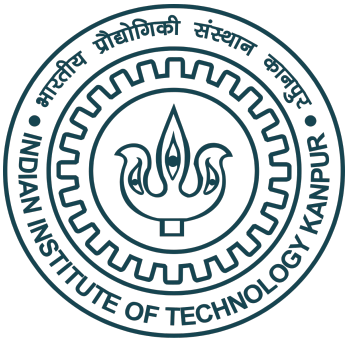
Experimental Measurement and Machine Learning-Based Prediction of Heat Generation in Lithium-Ion Batteries for Electric Vehicle Applications

A Thesis Submitted
In Partial Fulfillment of the Requirements
for the Degree of ***Doctor of Philosophy***

Vijay Kumar Chauhan

Roll No.:21129277



to the

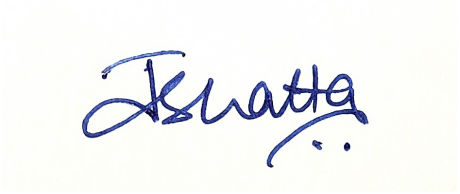
Department of Sustainable Energy Engineering,

Indian Institute of Technology Kanpur.

June, 2026

Certificate

It is certified that the work contained in the thesis titled **"Experimental Measurement and Machine Learning-Based Prediction of Heat Generation in Lithium-Ion Batteries for Electric Vehicle Applications"**, by **"Vijay Kumar Chauhan"** (**Roll No. 21129277**) has been carried out under my supervision and that this work has not been submitted elsewhere for a degree.



Prof. Jishnu Bhattacharya
(Supervisor)
Mechanical Engineering,
Indian Institute of Technology Kanpur.



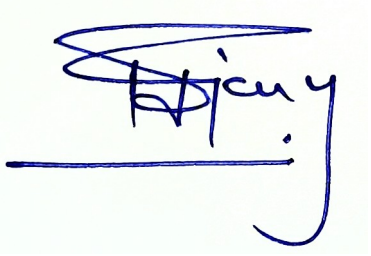
Prof. Vaibhav Arghode
(Co-Supervisor)
Sustainable Energy Engineering,
Indian Institute of Technology Kanpur.

June, 2026

Declaration

This is to certify that the thesis titled **"Experimental Measurement and Machine Learning-Based Prediction of Heat Generation in Lithium-Ion Batteries for Electric Vehicle Applications"** has been authored by me. It presents the research conducted by me under the supervision of **Prof. Jishnu Bhattacharya** and co-supervision of **Prof. Vaibhav Arghode**.

To the best of my knowledge, it is an original work, both in terms of research content and narrative, and has not been submitted elsewhere, in part or in full, for a degree. Further, due credit has been attributed to the relevant state-of-the-art and collaborations with appropriate citations and acknowledgments, in line with established norms and practices.



Vijay Kumar Chauhan
Roll No. 21129277
Doctor of Philosophy
Sustainable Energy Engineering,
Indian Institute of Technology Kanpur,
Kanpur, 208016, India.

Synopsis

Name of the Student: Vijay Kumar Chauhan **Roll No**: 21129277
Degree for which submitted: Ph.D.
Department: Department of Sustainable Energy Engineering
Thesis Title: Experimental Measurement and Machine Learning-Based Prediction of Heat Generation in Lithium-Ion Batteries for Electric Vehicle Applications
Thesis Supervisors: Prof. Jishnu Bhattacharya & Prof. Vaibhav Arghode
Month and Year of Thesis Submission: June 2026

Lithium-ion batteries have emerged as the dominant energy storage technology for electric vehicles (EVs) owing to their high energy density, long cycle life, and superior electrochemical performance. However, the safety, reliability, and performance of EV batteries are strongly influenced by their thermal behaviour, which is governed by the rate of heat generation within the cells. During real-world operation, EV batteries are subjected to highly dynamic conditions involving varying current loads, ambient temperatures, and states of charge. Accurate estimation of heat generation under such conditions is therefore essential for the design of efficient battery thermal management systems (BTMS), enhancement of battery safety, and improvement of overall vehicle performance. Despite extensive research in this area, existing heat generation estimation approaches are largely based on simplified assumptions and are often limited to constant-current operating conditions, making them inadequate for realistic EV applications.

Motivated by these challenges, the present research aims to develop a comprehensive framework for accurate heat generation estimation in lithium-ion batteries through the integration of electrochemical–thermal modelling, direct calorimetric measurements, and machine learning-based predictive approaches.

The first part of the study investigates the applicability of the conventional Bernardi equation under realistic EV drive cycle operation. A coupled electrochemical–thermal model is developed for a commercial 18650 lithium iron phosphate (LFP) cell and parameterized using experimental voltage data obtained under the Federal Urban Driving Schedule (FUDS) drive cycle. The fidelity of the model is subsequently validated using the US06 drive cycle.For both drive cycles, the model demonstrates high accuracy, with a root mean squared percentage error (RMSPE) of approximately 2%. The analysis reveals that the Bernardi equation overestimates the heat generation rate by approximately 25% under dynamic operating conditions. It is further observed that the reversible heat component becomes negligible during drive cycle operation, with heat generation being predominantly governed by irreversible processes. The developed electrochemical–thermal model accurately predicts both state of charge and temperature evolution, with prediction errors remaining within

7% and 2 °C, respectively. These findings demonstrate the limitations of conventional Bernardi-based heat estimation under realistic drive cycle operation and highlight the importance of experimentally validated electrochemical–thermal models for accurate thermal analysis. However, the high computational cost associated with such models limits their direct implementation in real-time EV applications, thereby motivating the development of computationally efficient data-driven approaches for practical battery thermal management.

The second part of the research focuses on the direct experimental characterization of heat generation behaviour using an isothermal battery calorimeter. Three commercially available 18650 lithium-ion cell chemistries, namely lithium iron phosphate (LFP), lithium nickel manganese cobalt oxide (NMC), and lithium cobalt oxide (LCO), are investigated over a wide operating range of discharge rates (0.5C–8C) and temperatures (10–60 °C). The results show that heat generation increases with increasing discharge rate and decreasing operating temperature across most operating conditions. Significant differences in thermal behaviour are observed among the investigated chemistries, with LFP and NMC cells exhibiting superior thermal adaptability and lower thermal losses compared to LCO cells. The study also reveals distinct reversible heat characteristics, including endothermic behaviour under specific operating conditions for LFP and NMC chemistries.

To quantitatively assess thermal performance, a novel metric termed Cell Discharge Efficiency (CDE) is introduced. The proposed metric accounts for both useful electrical energy output and thermal energy losses, enabling a comprehensive evaluation of battery thermal efficiency. Based on the CDE analysis, chemistry-specific operating windows are identified, providing practical guidelines for selecting thermally efficient operating conditions and battery chemistries for different applications. The developed framework offers valuable insights for BTMS design, battery selection, and thermal optimization in electric vehicle systems.

The final part of the research develops data-driven predictive models for accurate heat generation rate estimation. High-fidelity calorimetric data generated under diverse operating conditions are utilized to train and evaluate multiple machine learning algorithms, including Feedforward Neural Networks (FNN), Long Short-Term Memory (LSTM), Support Vector Machine (SVM), Random Forest Regressor (RFR), k-Nearest Neighbors (KNN) and Gaussian Process Regression (GPR). Feature importance analysis confirms that current and C-rate are the dominant parameters influencing heat generation behaviour, followed by depth of discharge and temperature. Among the investigated models, SVM demonstrates the highest predictive accuracy for both LFP and NMC chemistries, achieving coefficient of determination (R^2) values exceeding 0.999. To improve interpretability and facilitate real-time implementation, symbolic regression using the PySR framework is employed to derive compact analytical correlations for heat generation rate prediction. The resulting

expressions provide accurate and computationally efficient estimation of heat generation directly from measurable operating parameters, making them suitable for integration into battery management and thermal management systems.

Overall, the present research establishes a comprehensive experimental, modelling, and data-driven framework for heat generation estimation in lithium-ion batteries under realistic operating conditions. The work identifies the limitations of existing heat estimation approaches, provides detailed thermal characterization of major lithium-ion chemistries, introduces a novel thermal performance metric for operating window identification, and develops accurate machine learning-based predictive tools for real-time applications. The outcomes of this study contribute significantly to the advancement of battery thermal management strategies, thereby supporting the development of safer, more efficient, and thermally reliable electric vehicle battery systems.